

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.35	Gender	Male
Height (cm)	175.0	Weight (kg)	70.0
BMI	22.9	Product Info 2	A1
Occupation	Software Developer	Annual Income	\$120,000.00
Insurance Type	Life	Coverage Amount	\$500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Application ID: [N/A - Current Application]

Date: October 26, 2023

1. Risk Assessment

The applicant presents a profile that is generally favorable across most key risk indicators, with one critical exception:

* **Age:** The reported age of **0.35 years** is a significant anomaly. This implies an infant, which is highly inconsistent with other application details such as 'occupation' (Software Developer) and 'income' (\$120,000). This inconsistency requires immediate clarification. *Assuming this is a data entry error and should represent an adult age (e.g., 35 years), the following assessment is made under that premise, but the discrepancy remains a primary concern.*

* **BMI:** At 22.9 (Height: 175cm, Weight: 70kg), the applicant's BMI falls within the healthy range, indicating a low risk factor associated with weight.

* **Smoking Status:** Non-smoker (0). This is a highly favorable indicator, significantly reducing health risks.

* **Occupation:** 'Software Developer' is typically considered a low-to-moderate risk occupation, often associated with a sedentary lifestyle but generally stable employment and income.

* **Income:** An income of \$120,000 supports the requested coverage amount.

* **Previous Claims:** 0. No adverse claims history, which is favorable.

* **Family History:** 0. No adverse family health history reported, which is favorable.

* **Lifestyle:** Reports regular exercise (1) and no or minimal alcohol consumption (0), both contributing to a healthy lifestyle and lower risk.

* **Coverage Amount:** The requested coverage of \$500,000 represents approximately 4.17 times the annual income (\$120,000), which is a reasonable and common multiple for life insurance.

Overall, aside from the critical age anomaly, the applicant exhibits a low-risk profile across health and lifestyle factors.

2. Comparative Analysis

The Machine Learning Classifier outputs a Predicted Risk Class of 1 and a Calculated Annual Premium of \$10,000, aligning perfectly with two of the top similar historical approved cases:

* **Current Application:** Predicted Risk Class 1, Premium \$10,000.

* **POL-33998 (Customer_7):** Similarity: 100.0%, Risk Class: 1, Premium: \$10,000.00, Status: approved.

* **POL-49239 (Customer_9):** Similarity: 100.0%, Risk Class: 1, Premium: \$10,000.00, Status: approved.

The current application's predicted risk class and premium are identical to two previously approved cases with 100% feature similarity. This provides strong external validation for the ML model's assessment. The third similar case, POL-8902, was Risk Class 2 with a higher premium (\$12,000) and was also approved, further indicating that similar profiles have been deemed acceptable risks.

3. Anomaly & Fraud Detection

The primary and most critical anomaly detected in this application is the reported **age of 0.35 years**.

* This age is logically inconsistent with the applicant's stated 'occupation' as a 'Software Developer' and reported 'income' of \$120,000.

* Such a significant data discrepancy could be a data entry error, a system glitch, or potentially indicative of misinformation. This inconsistency renders the application un-underwritable in its current state.

No other significant anomalies were detected:

* The BMI (22.9) is healthy.

* The coverage amount (\$500,000) is reasonable relative to the stated income (\$120,000).

* Smoking, claims history, family history, exercise, and alcohol indicators are all favorable and consistent with a low-risk profile.

4. Premium Recommendation

The calculated annual premium of **\$10,000.00** is recommended *contingent upon the resolution of the age anomaly*.

This recommendation is based on:

* **Machine Learning Output:** The model predicts a favorable Risk Class 1, typically associated with lower premiums.

* **Historical Validation:** The predicted premium precisely matches two historical approved cases (POL-33998, POL-49239) with 100% similarity and an identical Risk Class 1.

* **Risk Profile (Adjusted for Age):** Assuming the age is corrected to an adult value (e.g., 35 years), the applicant's profile (healthy BMI, non-smoker, no adverse history, low-risk occupation, good lifestyle) aligns with a preferred or standard-plus risk classification, justifying a competitive premium.

* **Prediction Confidence:** The confidence level of 60.3% is moderate but, when combined with the strong historical comparables, supports the premium calculation.

5. Underwriting Decision Recommendation

Refer for Manual Review

Due to the critical and unexplainable discrepancy in the applicant's `age` (0.35 years) conflicting with their stated `occupation` and `income`, this application cannot proceed to automated approval or rejection.

Action Required:

1. **Clarify Applicant Age:** The primary step is to verify and correct the applicant's actual age.
2. **Re-evaluate:** Once the correct age is obtained, the application should be re-run through the ML classifier and re-evaluated against current underwriting guidelines.

If the age anomaly is resolved and the applicant is confirmed to be an adult with the otherwise favorable profile provided, an **approval** at the recommended premium of \$10,000 would be highly probable given the low risk class and strong historical comparables.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.